

Dr. Georgii Oblapenko

Born on 10 February 1992

Chair of Applied & Computational Mathematics (ACoM), RWTH Aachen, Schinkelstraße 2, 52062 Aachen, Germany



+49-0174-476-9489



oblapenko@acom.rwth-aachen.de



LinkedIn: George-Oblapenko



ResearchGate: Georgii-Oblapenko



ORCID: 0000-0002-1434-7166



Google Scholar: Georgii Oblapenko



Languages: English (fluent), German (fluent, including technical German), Russian (native speaker)

Researcher with 10+ years of experience in the field of numerical model development for simulation of non-equilibrium flows.

Employment history

Nov. 2023 – Present



Member of Scientific Staff • RWTH Aachen, Chair of Applied and Computational Mathematics.

Development of kinetic theory approaches for non-equilibrium flow simulations. Development of entropy-stable DG schemes for multi-component real gas flows with thermo-chemical non-equilibrium. Development and teaching of own Master course "Particle-based Simulation Methods". Teaching of Global Exercises in Math for bachelor students in the Computational Engineering and Math study program.

Jan. 2023 – Sep. 2023



Scientific Consultant • UT Austin, Oden Institute for Computational Engineering and Sciences / Department of Department of Aerospace Engineering and Engineering Mechanics.

Development of new PIC-DSMC collision algorithms. Mentoring of Master student developing PIC-DSMC methods for high-voltage gap closure modeling.

Nov. 2021 – Oct. 2023



Postdoctoral Researcher • German Aerospace Center (DLR), Institute of Aerodynamics and Flow Technology, Spacecraft Department.

Uncertainty quantification of hypersonic expanding continuum flows. Development of new Direct Simulation Monte Carlo methods for plasma simulation. Development of a photon Monte Carlo radiation transport code. Development of moment method models for gases with internal degrees of freedom. Work on low-rank methods for rarefied gas dynamic applications. Supported by a personal grant of the Alexander von Humboldt Foundation.

Employment history (continued)

- July 2018 – June 2021  **Postdoctoral Research Fellow** • UT Austin, Oden Institute for Computational Engineering and Sciences / Department of Department of Aerospace Engineering and Engineering Mechanics.
Development of a hybrid Discrete Velocity Method/Direct Simulation Monte Carlo code. Development of new Direct Simulation Monte Carlo algorithms for modeling of ionized rarefied gas flows. Development of new kinetic models for gases with internal degrees of freedom and chemically reacting gas flows. Research project supported by Sandia National Laboratories.
- Sept. 2015 – Dec. 2017  **Research engineer** • Saint-Petersburg State University, Department of Hydroaeromechanics.
Lead developer of a C++ library aimed for kinetic theory computations and coupling with CFD codes. Implementation of state-to-state models in Direct Simulation Monte Carlo (DSMC) codes, development of simplified models for vibrational relaxation rates, and numerical modeling of reaction rates in viscous gas flows.
- Apr. 2013 – Aug. 2015  **Research engineer** • Saint-Petersburg State University, Department of Hydroaeromechanics.
Development of theoretical models of reaction rates in viscous gas flows and their numerical modeling.

Education

- 2015 – 2017  **Ph.D., Saint-Petersburg State University** in Physical and Mathematical Sciences.
Area of specialization: Mechanics of Fluids, Gases and Plasma.
Thesis title: *Physico-chemical relaxation rates in viscous non-equilibrium gas flows*.
Supervisor: Prof. Kustova E.V.
- 2013 – 2015  **M.Sc. (summa cum laude), Saint-Petersburg State University** in Mechanics and Mathematical Modeling.
Area of specialization: Molecular-kinetic theory of fluids and gases.
Thesis title: *Vibrational relaxation rate in viscous rarefied gas flows*.
Supervisor: Prof. Kustova E.V.
- 2009 – 2013  **B.Sc., Saint-Petersburg State University** in Mathematics and Mechanics.
Supervisor: Prof. Kustova E.V.

Awards and personal funding

Personal funding and travel grants

- 2021 – 2023  **Alexander von Humboldt Foundation** Research Stipend for Postdoctoral Researchers (60000 EUR).
- 2017 – 2018  **German Academic Exchange Service (DAAD)** Research Stipend (3300 EUR).
- 2016  **Santander Bank** Research Stipend (1400 EUR).
- 2015 – 2016  **Russian Foundation for Basic Research** Research Stipend (4500 EUR).
- 2013, 2014, 2017  **Saint-Petersburg State University** Travel funding awards.

Awards

- 2023  **Marie Skłodowska-Curie Postdoctoral Action Seal of Excellence**

Awards and personal funding (continued)

- | | | |
|------|---|--|
| 2017 | ■ | Stipend of the Russian President for students and PhD students studying disciplines corresponding to prioritized areas of modernization of Russian economy (3800 EUR). |
| 2016 | ■ | Stipend of the Russian Government for students and PhD students studying disciplines corresponding to prioritized areas of modernization of Russian economy (1900 EUR). |
| | ■ | Stipend of the Russian Government for students and PhD students (500 EUR). |
| 2013 | ■ | Winner of the Saint-Petersburg Government grant competition for students and graduate students (500 EUR). |

Publications

The full list of publications consists of **23** peer-reviewed publications in SCOPUS/Web of Science-indexed journals and conference proceedings, as well as **16** publications without peer review (conference proceedings, preprints).

Statistics according to **SCOPUS** (March 2025):

- Number of publications: 27
- h-index: 10
- Citations: 323

Statistics according to **Google Scholar** (March 2025):

- Number of publications: 53
- h-index: 13
- Citations: 540

Conference and seminar talks

- | | | |
|-----------------------|---|---|
| Conferences | ■ | Presented at 23 international and 11 national (German, Russian) conferences. |
| Plenary talks | ■ | 2 invited plenary lecture at international conferences and workshops |
| Invited seminar talks | ■ | 21 invited talks at universities and research institutes in the USA, Germany, France, and the Netherlands. |

Teaching and student supervision

Teaching

- | | | |
|---------------------------|---|--|
| Summer semester 2024/2025 | ■ | Development and teaching of new Master course on “Particle-based Simulation Methods”, RWTH Aachen. |
| Winter semester 2024/2025 | ■ | Invited lecture on kinetic theory in the framework of the Master course on “Aerospace Fundamentals”, Justus Liebig University Giessen. |
| Summer semester 2023/2024 | ■ | Global Exercise for “Foundations of Mathematics II”, Bachelor Study Program “Computational Engineering Science”, RWTH Aachen. |

Teaching and student supervision (continued)

- Winter semester 2023/2024  Global Exercise for “Foundations of Mathematics I”, Bachelor Study Program “Computational Engineering Science”, RWTH Aachen.
Invited lecture on kinetic theory in the framework of the Master course on “Aerospace Fundamentals”, Justus Liebig University Giessen.

Additional teaching qualifications

- 2024–2025  Completed “Academic Teaching”, “Lecture Planning”, “Basics of RWTH-Moodle”, “Fundamentals of Examination”, and “Media Competency in Teaching” courses at Center for Teaching and Learning Services, RWTH Aachen.

Supervision, thesis reviews

- 2025  Supervision of a Bachelor thesis, RWTH Aachen.
Supervision of a Bachelor student research project at RWTH Aachen.
- 2024  Supervision of a Bachelor student research project at RWTH Aachen.
- 2022 – 2023  Supervision of a Master thesis at UT Austin.
- 2021  Referee of Master Thesis at Peter the Great St. Petersburg Polytechnic University .
- 2020 – 2021  Referee of 1 Bachelor-, 2 Diploma- and 2 Master theses at the Saint-Petersburg State University.

Professional membership and other activities

- Mar. 2022 – Mar. 2024  **COST Network** • Member of the “Mathematical models for interacting dynamics on networks” (MAT-DYN-NET) networking group, supported by the COST (European Cooperation in Science and Technology) program.
- Jul. 2021 – **Present**  **RGD NextGen Group** • Member of the RGD NextGen Group, aimed at increasing outreach of the Rarefied Gas Dynamics community.
- Spring 2021 – Jan, 2024  **American Institute of Aeronautics and Astronautics (AIAA) Thermophysics Technical Committee** • Member of Education, Emerging Technologies, Conference subcommittees (Spring 2021–January 2024).
- Dec. 2019 – Jan. 2024  **American Institute of Aeronautics and Astronautics (AIAA)** • Senior AIAA Member.

Editing and review activities

- Journal editor  Proceedings of the 33rd International Symposium on Rarefied Gas Dynamics, Springer, 2025 (co-editor)

Professional membership and other activities (continued)

Journal Reviewer  Computers and Fluids (Elsevier) · Acta Astronautica (Elsevier) · Journal of Computational Physics (Elsevier) · Computer Methods in Applied Mechanics and Engineering (Elsevier) · Applied Mathematical Modelling (Elsevier) · Physics of Fluids (AIP) · Journal of Thermophysics and Heat Transfer (AIAA) · Journal on Scientific Computing (SIAM) · Entropy (MDPI) · Fluids (MDPI) · Sensors (MDPI).

Proceedings Reviewer  International Symposium on Rarefied Gas Dynamics · STAB Symposium · AIAA Scitech · AIAA Aviation

Event organisation

RGD33 Symposium  Member of the Local Organizing Committee of the “33rd International Symposium on Rarefied Gas Dynamics”, 15-19 July 2024, Göttingen, Germany.

VIII Polyakhov Readings  Member of the Local Organizing Committee of the international conference on mechanics “VIII Polyakhov Readings”, 30 January-2 February 2018, Saint-Petersburg, Russia.

Hobbies

-  Music production
-  Painting
-  Running